# AI651 Assignment 2 Report

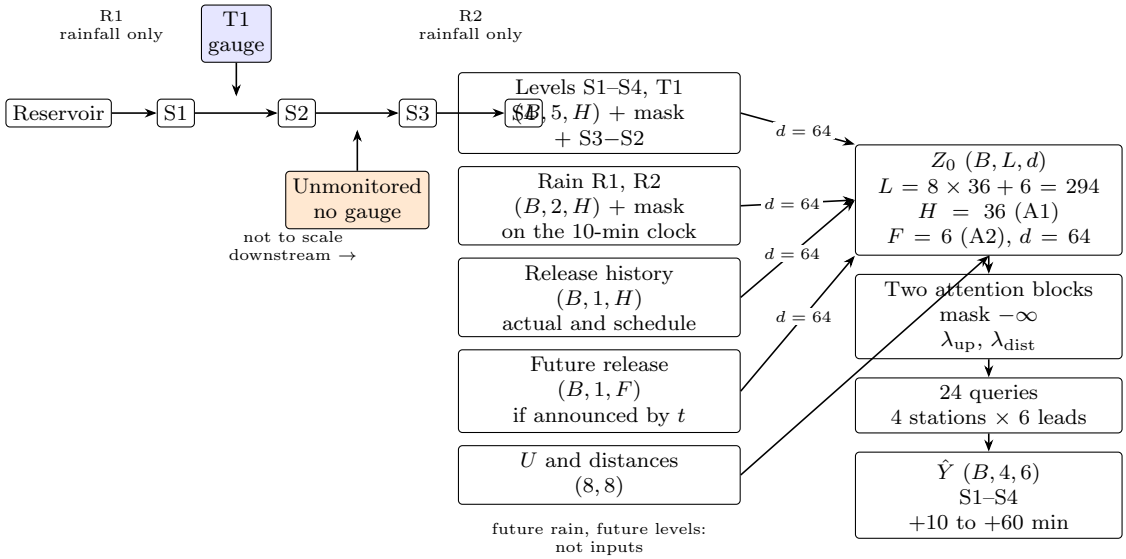
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**GitHub repository (public):** `https://github.com/ttqureshi/DL4STG-PA2`

## 1 Question 1

### 1.1 Architecture diagram




River, not to scale, and the forecast model. Each source has its own projection to width $d = 64$ before the concatenation. $H = 36$ and $F = 6$ are assumptions A1 and A2.

The eight series are projected and concatenated, and $\hat{Y}$ is the six future levels at S1–S4. $H = 36$, $F = 6$, and $d = 64$ are choices in assumptions A1 and A2.

### 1.2 Assumptions

The brief leaves the history length and the shape of the hour-ahead output unspecified, so A1 and A2 state them.

- **A1.** History is $H = 36$ bins, six hours, ending at issuance.

- **A2.** The output is six levels at each of S1–S4, at +10 through +60 minutes, so $\hat{Y}$ has shape $(B, 4, 6)$.
- **A3.** The archive covers at least one year, including wet and dry periods.
- **A4.** On a normal issuance the bin that closes at $t$ has arrived.
- **A5.** Gauge errors are small beside a rise that matters.
- **A6.** Each rain reading goes in the 10-minute bin that contains its timestamp, and a carried bin gets no attention weight.
- **A7.** Releases are piecewise constant between effective times, and the latest already-public announcement is used.
- **A8.** S1–S4 and T1 share 10-minute boundaries.
- **A9.** Water moves downstream, and travel time is not distance divided by a constant speed.
- **A10.** The target is level, and each series is standardized by its training mean and standard deviation.
- **A11.** R1 and R2 are not assigned to reaches; both are tokens, with gauge-to-station distance as a feature.
- **A12.** The unmonitored tributary has no gauge, and the S3−S2 residual is an extra history channel, zero on the other series.
- **A13.** Future rainfall and future levels are not inputs, and a future release is written only when it was announced by issuance.
- **A14.** Training uses the same clock: inputs at $t$ obey A13, and the target is the level at $t + \Delta, \ldots, t + 6\Delta$.
- **A15.** The loss is mean squared error of the standardized level over stations, leads, and the batch, with missing targets left out.
- **A16.** Along-river distance is used among the reservoir, S1–S4, and T1, and gauge-to-station distance is straight-line.

### 1.3 Task and data table

An issuance runs when a 10-minute bin closes at time $t$. The history is the $H = 36$ bins from $t - 35\Delta$ to $t$, and the outputs are the six levels at S1–S4 for leads $h = 1, \ldots, 6$. T1 is an input, and the batch size in operation is 1.

Inputs and the forecast at issuance $t$. Shapes are after alignment and before the width-$d$ projection. Masks are in $\{0, 1\}$.

| Name | Shape | Time | At issuance |
|---|---|---|---|
| Level history | $(B, 5, H)$ | $t-(H-1)\Delta \dots t$ | Known |
| Level mask | $(B, 5, H)$ | same bins | Known |
| S3−S2 residual | $(B, 5, H)$ | same; 0 if either missing | Constructed |
| Rain, aligned | $(B, 2, H)$ | level-clock bins | Past only |
| Rain mask | $(B, 2, H)$ | 1 if a real reading | Known |
| Actual release | $(B, 1, H)$ | bins ending at $t$ | Known |
| Schedule on history | $(B, 1, H)$ | already public | Known |
| Future release | $(B, 1, F)$ | $t+\Delta \dots t+6\Delta$ | If announced by $t$ |
| Future-release mask | $(B, 1, F)$ | 1 if public | Known |
| Upstream $U$, distances | $(8, 8)$ | static | Known |
| Forecast $\hat{Y}$ | $(B, 4, 6)$ | $t+\Delta \dots t+6\Delta$ | Target, never an input |

Series order is S1, S2, S3, S4, T1, R1, R2, release, so a shared lag index is the same 10-minute bin.

### 1.4 Notation, equations, and objective

$\Delta = 10$ minutes. Lag $\ell = 0, \dots, H-1$ means time $t-\ell\Delta$. Lead $h = 1, \dots, 6$ means time $t+h\Delta$. $U_{j,i} = 1$ when series $j$ is upstream of series $i$. Into S1: the release only. Into S2: the release, S1, and T1. Into S3: those plus S2. Into S4: those plus S3. A station is not upstream of itself. $\phi(d) = -\log(1 + d/d_0)$, with $d_0$ the median static distance. $\lambda_{\text{up}}$ and $\lambda_{\text{dist}}$ are learned scalars.

A level token carries the standardized level, the mask, and the residual channel. Rain carries standardized rain and its mask. A history release carries actual, schedule, and the actual mask. A future release carries the announced value and the known mask. Each group has its own matrix,

$$e = W_{\text{group}} x + a_i + b_{\text{time}},$$

and the token list is $Z^{(0)} \in \mathbb{R}^{L \times d}$. For block $n = 1, 2$,

$$\text{score}(p, q) = \frac{Q_p \cdot K_q}{\sqrt{d}} + \lambda_{\text{up}} U_{\text{series}(q),\text{series}(p)} + \lambda_{\text{dist}} \phi(d_{q,p}).$$

A key whose mask is 0 has score $-\infty$. The block is residual attention plus a residual feed-forward layer of width $4d$. The query for lead $h$ at station $s$ is $q_{s,h} = W_q[z_{s,0}; e_h; \tau_s]$, where $\tau_s$ is how many bins since $s$ last reported. The loss, with $m_{s,h}(t) = 1$ when the target was recorded, is the mean over the batch of

$$\frac{\sum_{s,h} m_{s,h} (\hat{y}^{\text{std}}_{s,h} - y^{\text{std}}_{s,h})^2}{\sum_{s,h} m_{s,h}}.$$

One attention block connects every remaining token, so S4 can see the reservoir in one step, and a second block reuses a mix the first has formed. With $L = 294$ and $d = 64$, one block is about $1.1 \times 10^7$ multiply-adds, and the ten-minute gap is set by when the gauges arrive. What is fixed is the series list, $U$, and the rule that an announcement enters only if it is already public. What is learned is every matrix, both blocks, the readout, and the four bias scalars.

### 1.5 Operation

**Missing observations.** A missing value is set to 0 after standardization and its mask to 0, so that token gets score $-\infty$.

**Delayed observations.** A bin that has not arrived by $t$ is missing, and the readout sees $\tau_s$, the bins since that station last reported.

**Announced future releases.** The latest announcement with announcement time at or before $t$ and effective time in the lead bin has known mask 1, and if none exists the mask is 0.

**Scheduled versus actual releases.** History bins carry the recorded actual and the announced schedule, and the future channel is only the schedule.

Each issuance is one forward pass of two attention blocks over 294 tokens of width 64 and 24 queries, which fits between the ten-minute bins.

### 1.6 Uncertainties and limitations

The unmonitored tributary can raise S3 and S4 with no trace in S2, T1, the release, or R1 and R2 until that water reaches S3. I use the S3−S2 residual and S3's history, and S4 reads those tokens in one attention block.

A changing travel time can put a peak in the wrong 10-minute bin. The model attends over every lag in the $H = 36$ window, with a separate query for each of the six leads, and uses distance as a feature.

A gauge on the unmonitored tributary would be a ninth series and would show that inflow before it reaches S3. Travel times at a low stage and a high stage would check whether $H = 36$ cuts off ordinary waves, and a catchment plus an error bar would fix which gauge belongs to which reach and how the loss weights the instruments. S1–S4 are still forecast without those extras.

## 2 Question 2

Write-up frame $f$ is tensor index $f - 1$. Every learned-model entry is the mean of seeds 0 and 1, with the two seeds in parentheses. Blank and Copy-last are one number each. Test numbers use videos 9000–9999, and every timed run is a Kaggle Tesla T4.

### 2.1 Table 1

The same mirror, rotation, and window functions are used in the experiments. Seed 0 gives $x = \text{torch.rand}(20, 1, 64, 64)$. The marker $p$ is zero except $p[7, 0, 10, 20] = 1$.

Table 1: Geometry and window checks. Maximum absolute difference is 0 and `torch.equal` is true for (a)–(c).

| Check | Result |
|---|---|
| (a) $\max \lvert M(M(x)) - x \rvert$ | 0, equal |
| (b) $\max \lvert R^4(x) - x \rvert$ | 0, equal |
| (c) $\max \lvert MRM(x) - R^{-1}(x) \rvert$ | 0, equal |
| (d) bright pixel of $M(p)$ | $(7, 10, 43)$ |
| (e) bright pixel of $R(p)$ | $(7, 43, 10)$ |
| (f) $1\times$, $t = 1$ | inputs 1–5, targets 6–10 |
| (g) $2\times$ | 1,3,5,7,9 / 11,13,15,17,19 |
| (h) $3\times$ | 1,4,7,10,13 / 16,19 |

## 2.2 Figure 1 and Q2.1–Q2.2

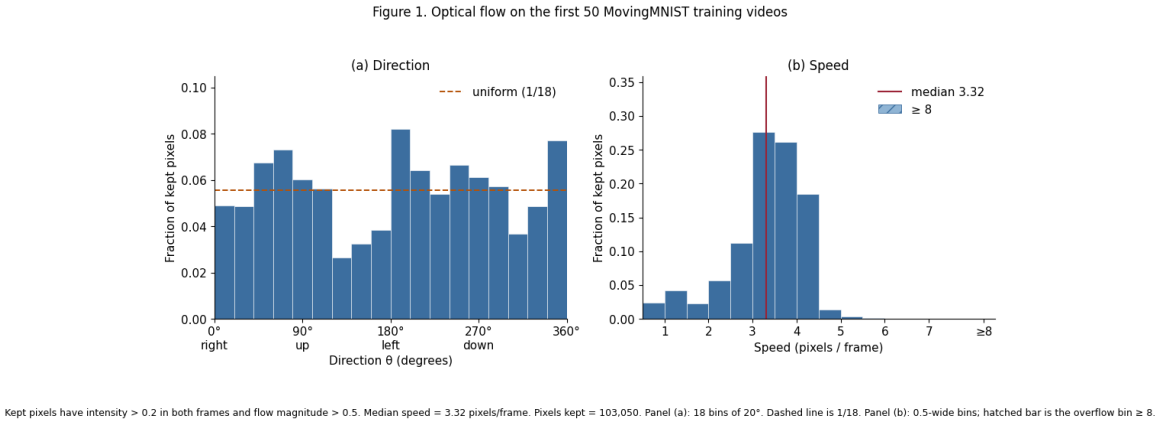



Figure 1: Farneback optical flow on the first 50 training videos, 950 frame pairs. Pixels kept have intensity above 0.2 in both frames and flow magnitude above 0.5. Median speed is 3.32 pixels/frame. Pixels kept: 103,050. Panel (a): 18 bins of 20°, dashed line at 1/18. Panel (b): 0.5-wide bins; the hatched bar is the overflow at or above 8.

**Q2.1.** Every 20° bin is occupied, and the amounts are not equal. The tallest bin is $[180°, 200°)$ at 0.0820, then $[340°, 360°)$ at 0.0772 and $[60°, 80°)$ at 0.0731, and the shortest is $[120°, 140°)$ at 0.0263. A mirror moves 13.2% of the 103,050 kept pixels into a different bin, and a 90° rotation moves 20.0%. The median speed is 3.32 pixels/frame, with interquartile range 3.00 to 3.89, so $2\times$ and $3\times$ are 6.64 and 9.95; 0.835 of kept pixels lie between 2.5 and 4.5, the 99th percentile is 4.67, and 0.00465 exceed 5. Overlap can scatter a minority of Farneback vectors, too few to move that median.

**Q2.2.** Mirror changes the glyph and sends $\theta$ to $180° - \theta$, and the speed is unchanged. Rotation changes the glyph further and adds 90° to every flow angle, and the speed is unchanged. Speed-up keeps the original frames and multiplies the step between observed frames by two or three.

### 2.3 Model A, Table 2, Figure 2, Q2.3–Q2.4

Model A has 480,784 trainable parameters: layer 1 has 184,576, layer 2 has 295,168, and the head has 1,040. On the Tesla T4 a training epoch took 11.3 seconds, and inference on the 1,000 test clips took 0.461 seconds.

Table 2: Test error at $1\times$, $t = 1$. Model A is the mean (seed 0, seed 1).

| | Model A | Blank | Copy-last |
|---|---|---|---|
| $\mathrm{MSE}_1$ | 0.0169 (0.0174, 0.0164) | 0.0426 | 0.0492 |
| $\mathrm{MSE}_2$ | 0.0203 (0.0209, 0.0198) | 0.0426 | 0.0604 |
| $\mathrm{MSE}_3$ | 0.0226 (0.0231, 0.0221) | 0.0426 | 0.0669 |
| $\mathrm{MSE}_4$ | 0.0244 (0.0248, 0.0240) | 0.0425 | 0.0708 |
| $\mathrm{MSE}_5$ | 0.0261 (0.0264, 0.0257) | 0.0425 | 0.0732 |
| $E(5)$ | 0.0221 (0.0225, 0.0216) | 0.0426 | 0.0641 |
| Parameters | 480,784 | 0 | 0 |

Figure 2. Model A forecasts on three test videos

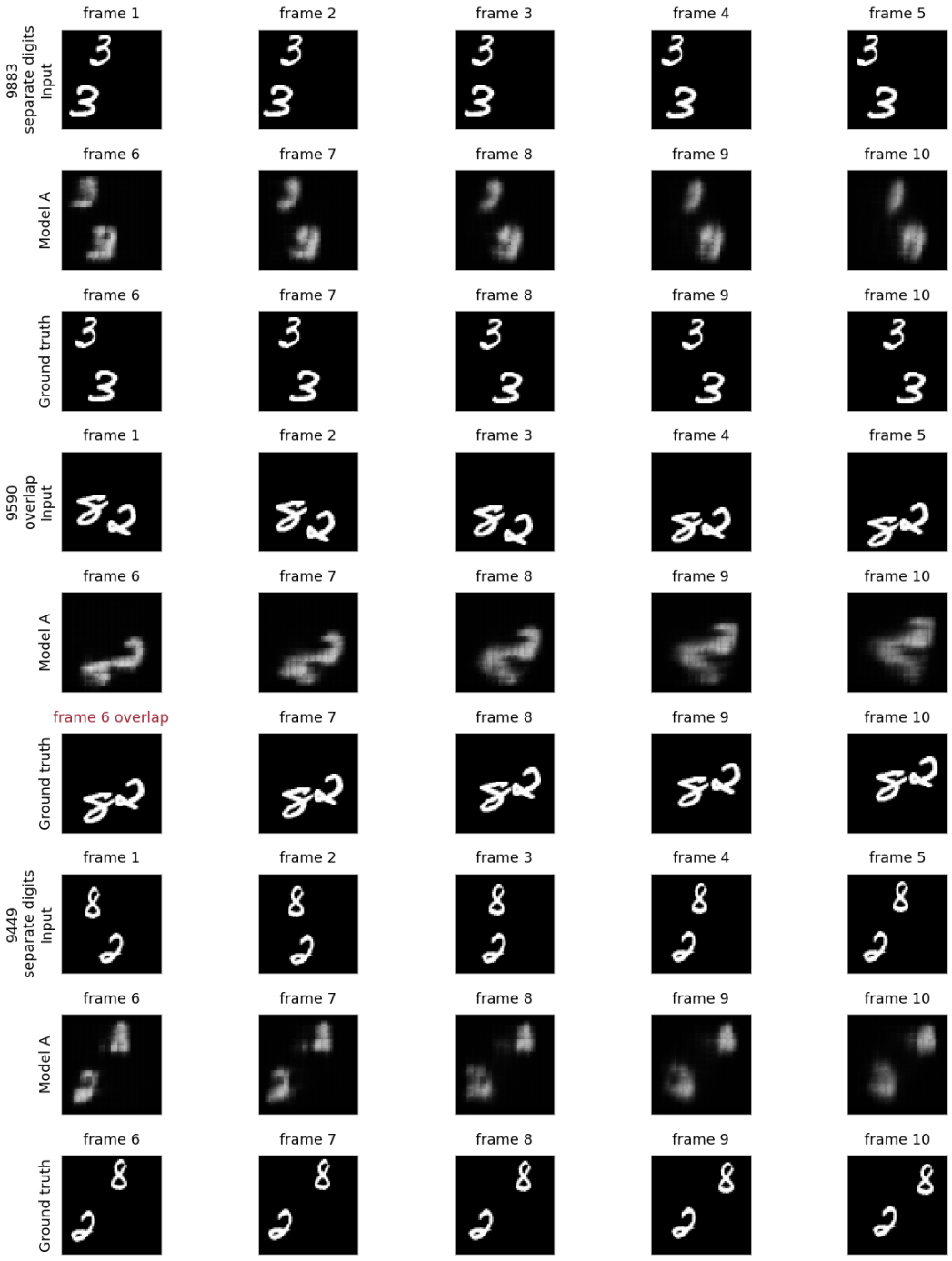



Figure 2. Model A, seed 0, on MovingMNIST test videos 9883, 9590, and 9449 (dataset indices). Each block shows input frames 1–5, the five predicted frames, and ground-truth frames 6–10. Index 9590 has overlapping digits in ground-truth frame 6.

Figure 2: Model A, seed 0, on test videos 9883, 9590, and 9449. Each block shows input frames 1–5, the five predicted frames, and ground-truth frames 6–10. Index 9590 has overlapping digits in ground-truth frame 6.

**Q2.3.** Model A's one-step reach is 8 original pixels. `pixel_unshuffle(4)` leaves a $16 \times 16$ grid whose neighbouring cells are 4 original pixels apart, centre to centre. The fold, the $1 \times 1$

head, and `pixel_shuffle` do not move information to another cell. Each $3 \times 3$ kernel moves one folded cell. Inside one step the new input passes through layer 1 and then through layer 2, which reads layer 1's new hidden state. That is two cells, $2 \times 4 = 8$ original pixels. The previous hidden state of layer 1 travels the same two kernels. The previous hidden state of layer 2 travels only one, which is 4 original pixels. The longest path is $u_\tau \to$ layer-1 $3 \times 3 \to h_\tau^{(1)} \to$ layer-2 $3 \times 3 \to \hat{x}_{\tau+1}$.

**Q2.4.** Model A's $E(5)$ of 0.0221 is below Blank (0.0426) and Copy-last (0.0641) at every horizon. The seed-mean error grows from 0.0169 at horizon 1 to 0.0261 at horizon 5, and horizon 5 is still below Blank. Copy-last leaves ink where the digit has left. On 9883 and 9449 the forecast is a soft mark near each separate digit; on 9590 the digits cross and the seed-0 forecast is still one smear at frames 9 and 10, where the truth has separated.

### 2.4 Q2.5

This answer was committed as `10cb5f4` at 14:35 on 27 October 2026, before the mirror, rotation, and speed tests. It is unchanged.

Model A will not pass the mirror test. Its one-step reach is 8 original pixels, which is a statement about how far a $3 \times 3$ convolution can move information, not about reflection. The filters are ordinary convolutions, with no mirrored copy of the weights. Figure 1(a) is not symmetric under a horizontal flip: reflecting the measured flow sends $\theta$ to $180^\circ - \theta$ and moves 13.2% of the 103,050 kept pixels into a different $20^\circ$ bin. A mirrored digit is also a different glyph. Both the direction balance and the appearance sit outside the $1\times$ training sample, so $f(M(X))$ should disagree with $M(f(X))$ and with $M(Y)$.

Model A will not pass the rotation test, and rotation should be worse than mirroring. A $90^\circ$ anticlockwise turn changes the glyph further than a left-right flip, and on these same pixels it reassigns 20.0% of the kept flow into a different bin. Nothing in the two-layer ConvLSTM ties a filter to its quarter-turn.

It should break at $3\times$, and $2\times$ should already be strained for the faster digits. The median consecutive-frame speed is 3.32 pixels per frame, inside the 8-pixel reach. Twice that median is 6.64, still under 8 but close. The 99th percentile at $1\times$ is 4.67, which becomes about 9.3 at $2\times$ and is already past the reach. Three times the median is 9.95, beyond 8.

### 2.5 Tables 3–4 and Figures 3–5

Table 3: Consistency and transformed error at $1\times$, $t = 1$. Blank and Copy-last have $C_M = C_R = 0$.

| | Model A | Blank | Copy-last |
|---|---|---|---|
| $E(5)$ | 0.0221 (0.0225, 0.0216) | 0.0426 | 0.0641 |
| $C_M$ | 0.00183 (0.00168, 0.00198) | 0 | 0 |
| $E_M$ | 0.0227 (0.0232, 0.0223) | 0.0426 | 0.0641 |
| $C_R$ | 0.00308 (0.00315, 0.00302) | 0 | 0 |
| $E_R$ | 0.0238 (0.0242, 0.0234) | 0.0426 | 0.0641 |

Figure 3. Seed-0 Model A, largest per-video consistency gaps




Figure 3. Seed-0 Model A at 1x, t=1. The two test videos with the largest per-video C_M are dataset indices 9347 (C_M=0.00624) and 9257 (C_M=0.00466). The two with the largest per-video C_R are 9118 (C_R=0.00729) and 9493 (C_R=0.00676). Each panel is four rows by ten frames: columns 1-5 are the input side and columns 6-10 are the future side. Row 1 is X | f(X), row 2 is T(X) | T(Y), row 3 is T(X) | f(T(X)), row 4 is T(X) | T(f(X)). GIFs keep those four rows and step through the five forecast horizons, with the five input frames held on the left and the current future frame on the right.
GIF paths: figures/gifs/figure3_cm_rank1.gif, figures/gifs/figure3_cm_rank2.gif, figures/gifs/figure3_cr_rank1.gif, figures/gifs/figure3_cr_rank2.gif.

Figure 3: Seed-0 Model A. Mirror ranks are test videos 9347 ($C_M = 0.00624$) and 9257 ($C_M = 0.00466$). Rotation ranks are 9118 ($C_R = 0.00729$) and 9493 ($C_R = 0.00676$). Each panel has four rows: $X \mid f(X)$; $T(X) \mid T(Y)$; $T(X) \mid f(T(X))$; $T(X) \mid T(f(X))$. GIFs: `figures/gifs/figure3_cm_rank1.gif`, `figures/gifs/figure3_cm_rank2.gif`, `figures/gifs/figure3_cr_rank1.gif`, `figures/gifs/figure3_cr_rank2.gif`.

Figure 4. Per-video consistency on the 1,000 test videos

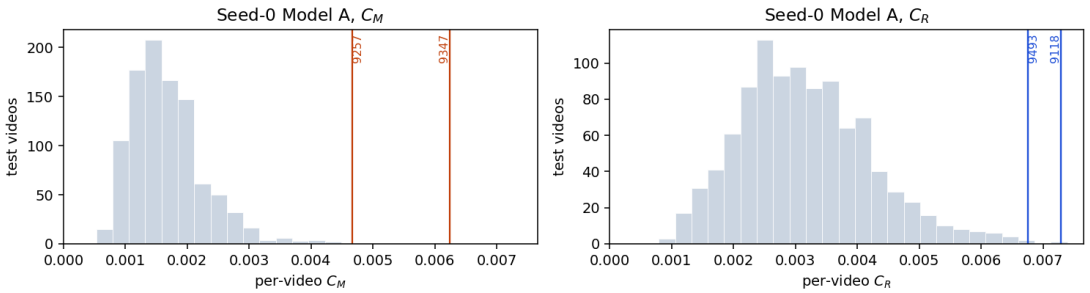



Figure 4. Histograms of per-video C_M and C_R for seed-0 Model A on the 1,000 test videos, dataset indices 9000-9999. Vertical lines mark the Figure 3 videos. C_M lines: 9347 (0.00624), 9257 (0.00466). C_R lines: 9118 (0.00729), 9493 (0.00676).

Figure 4: Per-video $C_M$ and $C_R$ of seed-0 Model A on the 1,000 test videos. The Figure 3 videos are marked.

Table 4: Speed comparison at $t = 1$. $E(5)$ at $3\times$ is omitted.

| | Model A | Copy-last |
|---|---|---|
| $E(2)$, $1\times$ | 0.0186 (0.0191, 0.0181) | 0.0548 |
| $E(2)$, $2\times$ | 0.0448 (0.0449, 0.0448) | 0.0661 |
| $E(2)$, $3\times$ | 0.0520 (0.0528, 0.0511) | 0.0703 |
| $E(5)$, $1\times$ | 0.0221 (0.0225, 0.0216) | 0.0641 |
| $E(5)$, $2\times$ | 0.0490 (0.0486, 0.0493) | 0.0719 |

Figure 5. Seed-0 Model A, test video 9883, speeds 1x 2x 3x

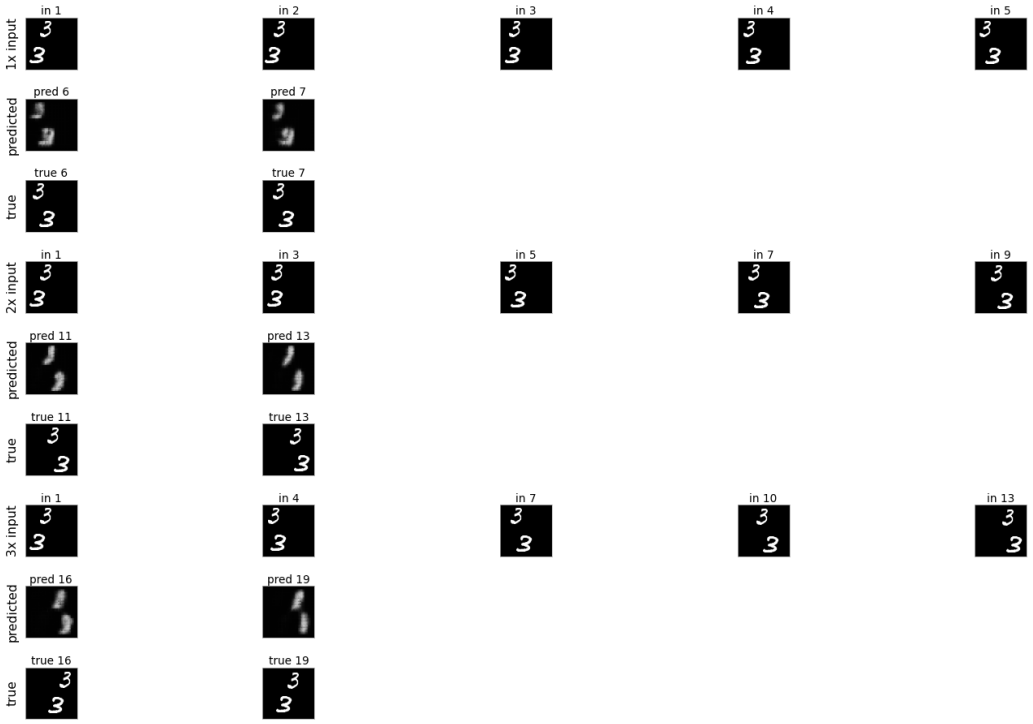



Figure 5. Seed-0 Model A on test video 9883. Each speed shows five inputs and the first two predicted and true futures at t=1. Non-overlap: every displayed true frame has exactly two 8-connected components of pixels above 0.1, each of area at least 80 (area>=80 and combined frame-5-to-7 displacement>=4). 35 videos qualified. Minimum centroid distance 31.19 px.

Figure 5: Seed-0 Model A on test video 9883. Five inputs and the first two predicted and true futures at 1×, 2×, and 3×. The displayed true frames keep the two digits apart.

**Q2.6.** Blank and Copy-last have $C_M = C_R = 0$, and their transformed errors equal their $E(5)$, 0.0426 and 0.0641. A zero gap can sit on a wrong forecast, so $C_T$ is reported next to $E_T$.

**Q2.7.** On video 9347, seed 0, row 3 still has two digits at horizon 5, with the second digit too far down and not far enough left, while row 4 is one blob. The per-video $C_M$ is 0.00624.

**Q2.8.** Rotation is harder than mirroring for Model A: $C_R = 0.00308$ against $C_M = 0.00183$, and $E_R = 0.0238$ against $E_M = 0.0227$. Rotation moves 20.0% of the kept flow against 13.2% for the mirror, and it changes the glyph further, while the speed is the same under both.

**Q2.9.** Model A falls apart at 2×, and 3× is worse: $E(2)$ goes from 0.0186 at 1× to 0.0448 at 2×, above Blank's 0.0426, and to 0.0520 at 3×. The 2× median step of 6.64 is inside the 8-pixel reach, so a step near that edge, and the tail past 8, breaks the forecast before the typical 3× step of 9.95. In Figure 5 the 1× forecast follows both digits, and at 2× and 3× it is one soft blob.

**Q2.10.** The baseline does not separate a missing fast example from a step beyond reach, because Model A never saw 2× or 3× and its reach is 8 pixels. Models C and D are that experiment: C keeps A's architecture and adds speeds $\{1, 2, 3\}$, and D keeps that schedule and lengthens the reach.

## 2.6 Q2.11

This answer was committed as `ede80b4` at 16:20 on 27 October 2026, before training B, C, and D. It is unchanged.

Model B should close the mirror and rotation gaps, the rotation gap most of all, because it trains on the group $G$, and it should not fix the speed failure because it never sees $2\times$ or $3\times$. Model C should improve $2\times$ if that error is unfamiliar motion inside the 8-pixel reach, and it should leave $3\times$ broken, because a typical $3\times$ step is 9.95 pixels and C still has Model A's reach. Model D, with reach 16 and the same speed schedule as C, is the one that should fix the $3\times$ failure, and the part of the $2\times$ failure that sits near the edge of 8 pixels.

## 2.7 Models B, C, and D

Model B is Model A trained at $1\times$ with one uniform draw from the eight elements of $G$ applied to the input and the target. Model C is Model A trained with $k$ drawn from $\{1, 2, 3\}$ and no symmetry augmentation. Model D is the same schedule as C, with dilation 2 and padding 2 on both $3 \times 3$ layers. Dilation does not add weights, so D also has 480,784 parameters, under the cap of 721,176. No attention is used.

**Q2.12.** Model D's one-step reach is 16 original pixels, against Model A's 8. A $3 \times 3$ kernel with dilation 2 reads offsets $-2, 0, +2$, so one layer moves 2 folded cells, which is 8 original pixels. The new input passes through both layers, and layer 2 reads layer 1's new hidden state, so the longest path is 4 folded cells, $4 \times 4 = 16$. The path is $u_\tau \to$ layer-1 dilated $3 \times 3 \to h_\tau^{(1)} \to$ layer-2 dilated $3 \times 3 \to \hat{x}_{\tau+1}$. The $1 \times 1$ head and the shuffles add nothing. The previous state of layer 2 travels only one dilated kernel, which is 8 original pixels.

Table 5: Test metrics for A–D. Consistency is at $1\times$. $E(2)$ at $3\times$ uses the first two horizons only. Training and inference times are seconds on a Tesla T4.

| | A | B | C | D |
|---|---|---|---|---|
| $E(5)$, $1\times$ | 0.0221 (0.0225, 0.0216) | 0.0223 (0.0226, 0.0221) | 0.0267 (0.0268, 0.0267) | 0.0287 (0.0283, 0.0290) |
| $C_M$ | 0.00183 (0.00168, 0.00198) | 0.00106 (0.00108, 0.00105) | $9.30 \times 10^{-4}$ | $9.81 \times 10^{-4}$ |
| $E_M$ | 0.0227 (0.0232, 0.0223) | 0.0224 (0.0227, 0.0221) | 0.0269 (0.0270, 0.0269) | 0.0289 (0.0286, 0.0292) |
| $C_R$ | 0.00308 (0.00315, 0.00302) | 0.00132 (0.00126, 0.00139) | 0.00205 (0.00188, 0.00222) | 0.00181 (0.00166, 0.00197) |
| $E_R$ | 0.0238 (0.0242, 0.0234) | 0.0223 (0.0226, 0.0219) | 0.0279 (0.0279, 0.0278) | 0.0296 (0.0292, 0.0299) |
| $E(2)$, $1\times$ | 0.0186 (0.0191, 0.0181) | 0.0186 (0.0187, 0.0184) | 0.0230 (0.0230, 0.0230) | 0.0263 (0.0261, 0.0266) |
| $E(2)$, $2\times$ | 0.0448 (0.0449, 0.0448) | 0.0446 (0.0444, 0.0448) | 0.0288 (0.0288, 0.0289) | 0.0293 (0.0289, 0.0297) |
| $E(2)$, $3\times$ | 0.0520 (0.0528, 0.0511) | 0.0506 (0.0496, 0.0515) | 0.0335 (0.0336, 0.0334) | 0.0316 (0.0311, 0.0321) |
| Parameters | 480,784 | 480,784 | 480,784 | 480,784 |
| s/epoch | 11.3 | 11.6 | 12.4 | 19.9 |
| Inference (s) | 0.461 | 0.464 | 0.453 | 0.720 |

Model C's seed pair for $C_M$ is $8.82 \times 10^{-4}$ and $9.77 \times 10^{-4}$. Model D's is $9.90 \times 10^{-4}$ and $9.71 \times 10^{-4}$.

**Q2.13.** The mirror and rotation gaps are appearance: Model B cuts $C_M$ from 0.00183 to 0.00106 and $C_R$ from 0.00308 to 0.00132, with $E(5)$ still 0.0223 and $E(2)$ still 0.0446 at $2\times$ and 0.0506 at $3\times$. The $2\times$ failure is unfamiliar motion: Model C, at the same 8-pixel reach, has $E(2) = 0.0288$ at $2\times$ against Model A's 0.0448 and Model B's 0.0446, and Model D is level with C at 0.0293, inside the 6.64-pixel typical step. The $3\times$ failure is that missing example and then reach: Model C falls from 0.0520 to 0.0335, and Model D, reach 16, is lower at 0.0316, while a typical $3\times$ step of 9.95 pixels is past 8. More $3\times$ windows cannot add the folded cells that dilation 2 inserts.

**Q2.14.** Model B's $E(5)$ at $1\times$ is 0.0223 against Model A's 0.0221, so the symmetry training did not cost the original clips, and it cut $C_M$ to 0.00106 and $C_R$ to 0.00132. Model C's $E(5)$

rises to 0.0267, a rise of 0.00467, while $E(2)$ falls by 0.0160 at 2× and by 0.0185 at 3×, and $E(2)$ at 1× rises from 0.0186 to 0.0230. Model D's $E(5)$ is 0.0287; against C the 1× cost is about 0.00193 and the 3× gain is about 0.00185, and at 2× D's 0.0293 does not beat C's 0.0288.

### 2.8 The final model

Table 5 supports D+B. Model B cut the rotation gap without moving $E(5)$, and Model D's rotation gap (0.00181) was still larger than its mirror gap. D+B is Model D trained with both the speed draw and one shared group element on each window. Its $E(5)$ is 0.0285 (0.0287, 0.0282), slightly under D. Its gaps are $C_M = 7.53 \times 10^{-4}$ and $C_R = 9.89 \times 10^{-4}$, with $E_M = E_R = 0.0285$. Its $E(2)$ is 0.0261, 0.0290, and 0.0313 at 1×, 2×, and 3×. The speed gains survive. The final model is that network wrapped as

$$\bar{f}(X) = \frac{1}{8} \sum_{g \in G} g^{-1}\big(f(g(X))\big).$$

For any $h \in G$, right multiplication by $h$ permutes $G$ and $h$ factors out of the average, so $\bar{f}(h(X)) = h(\bar{f}(X))$. Since $M$ and $R$ are in $G$, $C_M = C_R = 0$ exactly. The measured gaps are the floating-point residual.

Table 6: Symmetrised forecasters. Training time is the underlying model's seconds per epoch. Inference includes all eight transforms. $C_M$ and $C_R$ are in scientific notation.

| | Sym. B | Sym. D | Sym. D+B (final) |
|---|---|---|---|
| $E(5)$, 1× | 0.0218 | 0.0285 | 0.0281 (0.0284, 0.0279) |
| $C_M$ | $8.43 \times 10^{-17}$ | $5.47 \times 10^{-17}$ | $5.65 \times 10^{-17}$ |
| $E_M$ | 0.0218 | 0.0285 | 0.0281 |
| $C_R$ | $6.76 \times 10^{-17}$ | $4.26 \times 10^{-17}$ | $4.48 \times 10^{-17}$ |
| $E_R$ | 0.0218 | 0.0285 | 0.0281 |
| $E(2)$, 1× | 0.0181 | 0.0263 | 0.0259 (0.0261, 0.0257) |
| $E(2)$, 2× | 0.0438 | 0.0291 | 0.0286 (0.0288, 0.0285) |
| $E(2)$, 3× | 0.0495 | 0.0314 | 0.0309 (0.0311, 0.0306) |
| Parameters | 480,784 | 480,784 | 480,784 |
| s/epoch | 11.6 | 19.9 | 20.7 |
| Inference (s) | 3.81 | 5.24 | 4.95 |

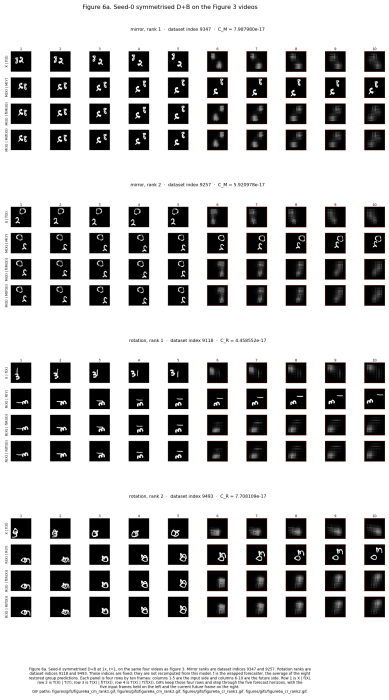



Figure 6b. Seed-0 symmetrised D+B, dataset 9883, speeds 1x 2x 3x

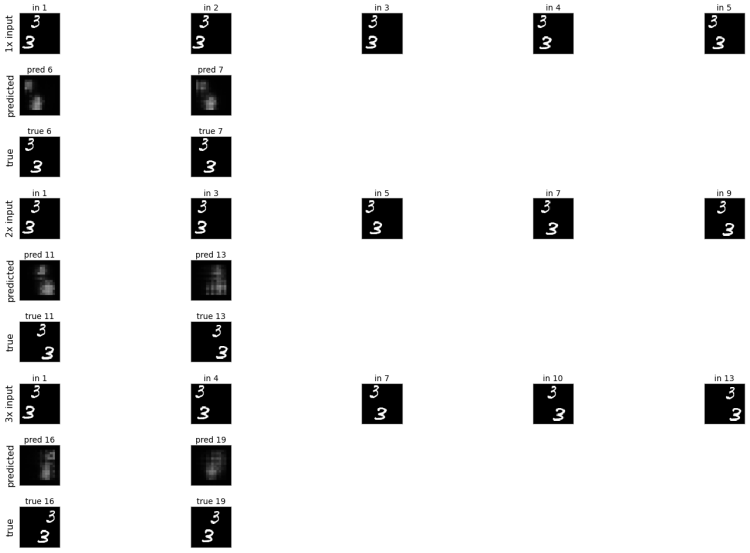



Figure 6b. Seed-0 symmetrised D+B on dataset index 9883, the same video as Figure 5. Each speed shows five inputs and the first two predicted and true futures at t=1. At 3x only those two futures exist inside the clip.

Figure 6: Symmetrised D+B, seed 0. (a) The Figure 3 videos: mirror 9347 and 9257, rotation 9118 and 9493. GIFs: `figures/gifs/figure6a_cm_rank1.gif`, `figures/gifs/figure6a_cm_rank2.gif`, `figures/gifs/figure6a_cr_rank1.gif`, `figures/gifs/figure6a_cr_rank2.gif`. (b) Test video 9883, the same video as Figure 5.

**Q2.15.** The wrapper is exact and the learned symmetry is not: Model B still has $C_M = 0.00106$ and $C_R = 0.00132$, and D+B has $7.53 \times 10^{-4}$ and $9.89 \times 10^{-4}$, while the wrapped gaps are of order $10^{-17}$ and $E_M = E_R = E(5)$. Wrapped B's $E(5)$ is 0.0218 against learned B's 0.0223, and wrapped D+B's is 0.0281 against 0.0285; the mean squared spread of the eight forecasts is $3.91 \times 10^{-4}$ (worst video $1.13 \times 10^{-3}$), so the average did not raise the test error.

Inference is the cost: wrapped B takes 3.81 s (4.42, 3.20) against learned B's 0.464 s (0.517, 0.412), wrapped D takes 5.24 s (5.20, 5.28) against 0.720 s (0.725, 0.715), and wrapped D+B takes 4.95 s (4.90, 5.00) against 0.712 s (0.718, 0.705), about 8.2, 7.3, and 7.0 times as long, with D+B's training time still 20.72 s per epoch (20.67, 20.78).

**Q2.16.** On video 9347, the video in Q2.7, symmetrised D+B makes rows 3 and 4 the same image: the per-video $C_M$ is $7.99 \times 10^{-17}$. Horizon 5 is still wrong. The truth has two digits and the forecast is one blob between them, because the underlying forecast had already merged and the wrapper copies that merge into both orders.

**Q2.17.** The final model learned the speeds that were in training, and it did not learn to keep two digits apart. Symmetrised D+B has mean $E(2)$ 0.0259 at 1×, 0.0286 at 2×, and 0.0309 at 3×, against Model A's 0.0186, 0.0448, and 0.0520, while Blank stays near 0.0426. On video 9883 the truth keeps two digits. The prediction is one component by frame 7 at 1× and by frame 11 at 2×, and at 3× it is one blob between the two paths. A skeptic should require both centroids to track the truth on a non-overlapping clip at a speed the training loop never drew. A 4× window has not been run.

## AI use disclosure

I used Cursor (Composer) while coding and while drafting this report.

**Coding prompts (summary):** set up the repo off Google Drive; run Table 1; check that Model A has 480,784 parameters; train on a Kaggle Tesla T4; make the optical-flow figure; design the river model; design Model D; train B, C, D, and D+B; symmetrise the forecaster; write the report; shorten the report and draw the Question 1 architecture.

**Model help on code:** wrote `transforms.py`, `windows.py`, `data.py`, `metrics.py`, `convlstm.py`, `model_a.py`, `model_d.py`, and `symmetrise.py`; wrote the training scripts `train_a.py`, `train_b.py`, `train_c.py`, `train_d.py`, and `train_db.py`; ran Table 1, Figure 1, and every training and evaluation job on Kaggle (Tesla T4). Model D's dilation and the choice to train D+B were made in that work.

**My edits on code:** checked that Table 1 matches, that Model A and Model D each have 480,784 parameters, that Q2.5 and Q2.11 were pushed before the experiments they precede, and that every reported test number matches the result JSON files.

**Question 1:** Cursor drafted the six headings. The history length and the six-step horizon are assumptions A1 and A2, because the brief leaves both unspecified. I checked that the section separates those assumptions from the given river, and that future rain is not an input while an already announced release is. The diagram is the TikZ figure, and the six headings were shortened.

**Question 2:** Cursor implemented the model, training, and figure scripts and ran them on Kaggle (Tesla T4). I checked Table 1, the 480,784 parameter count, and that the reported test numbers come from the result files. Q2.5 was committed as `10cb5f4` at 14:35 on 27 October 2026, before the mirror, rotation, and speed tests. Q2.11 was committed as `ede80b4` at 16:20 the same day, before training B, C, and D.